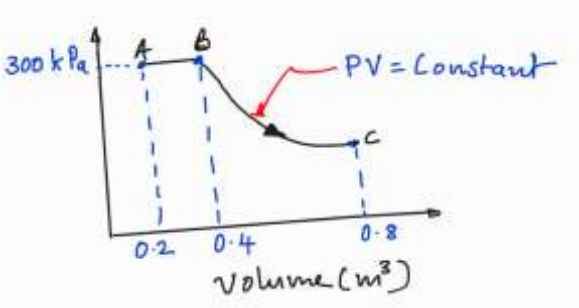
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF Aeronautical & Automobile Engineering			
END SEMESTER EXAMINATION/		**November, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**Aeronautical /Automobile Engineering**	**Course Type**	**Core**
Course Name	**Engineering Thermodynamics**	**Course Code**	**AAE 2123**
Semester	**III**	**Max Marks**	**50**
Date	**22/11/2024**	**Duration**	**180 Minutes**

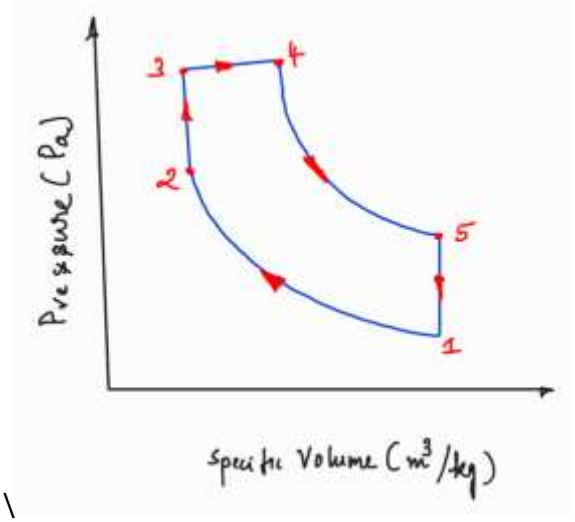
Instructions:

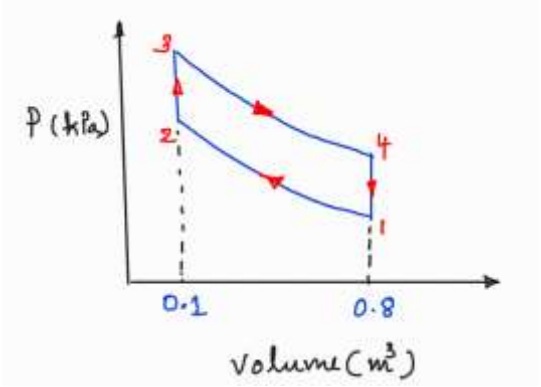
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET		BL
			CO	PO	CLO	AHEP4 LO	
1A	Show that work transfer in a polytrocpic process is given by $\boldsymbol{W = \frac{P_1V_1 - P_2V_2}{n-1}}$ A mass of gas is compressed in a quasi-static process from 80kPa and 0.1 m^3 to 0.4MPa and 0.03m^3. Assuming the Pressure-Volume relation as PV^n=Constant, find the work transfer in this case.	5	CO1	PO1, PO2, PO3, P012	CLO1	LO1, LO2, LO3	3
1B	Determine the work transfer in a process represented by the P-V diagram shown below.  **FIGURE 1**	3	CO1	PO1, PO2, PO3, P012	CLO1	LO1, LO2, LO3	4

<table>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>1C</td><td>What is the difference between open system and closed system. Give one example for each.</td><td>2</td><td>CO1</td><td>PO1, PO2, PO3, P012</td><td>CLO1</td><td>LO1, LO2, LO3</td><td>2</td></tr>
<tr><td>2A</td><td>A system undergoes a cycle during operation. The heat and work transfers in each processes are as tabulated. Complete the table. Given: The net work transfer in cycle is -17000kJ.
<table>
<tr><th>Process</th><th>Heat transfer (kJ)</th><th>Work transfer (kJ)</th><th>Change of Internal Enegry (kJ)</th></tr>
<tr><td>a-b</td><td>0</td><td>2170</td><td>_____</td></tr>
<tr><td>b-c</td><td>21000</td><td>0</td><td>_____</td></tr>
<tr><td>c-d</td><td>-2100</td><td>_____</td><td>-36600</td></tr>
<tr><td>d-a</td><td>_____</td><td>_____</td><td>_____</td></tr>
</table></td><td>4</td><td>CO2</td><td>PO1, PO2, PO3, P012</td><td>CLO2</td><td>LO1, LO2, LO3</td><td>4</td></tr>
<tr><td>2B</td><td>A reversible heat engine operating between two reservoir at temperature 400^0C and 40^0 C. This engine is used to drive a reversible refrigerator which transfers heat from -20^0C to 40^0C. The engine receives a heat of 2000kJ and combined system has net work output of 360kJ. Find
a) Heat removed from cold body by refrigerator
b) Heat received by 40^0C reservoir</td><td>4</td><td>CO2</td><td>PO1, PO2, PO3, P012</td><td>CLO2</td><td>LO1, LO2, LO3</td><td>3</td></tr>
<tr><td>2C</td><td>Give two statements of second law of Thermodynamics. Show a schematic.</td><td>2</td><td>CO2</td><td>PO1, PO2, PO3, P012</td><td>CLO2</td><td>LO1, LO2, LO3</td><td>2</td></tr>
<tr><td>3A</td><td>Water is heated at 3 bar till its temperature reaches to 200^0C. what is the condition of water?
If its is cooled in a constant volume , at what temperature will it become saturated steam? If the steam is cooled further to 80^0C, what is the final quality and enthalpy of steam.
Plot the states and process in T-v diagram.</td><td>4</td><td>CO3</td><td>PO1, PO2, PO3, P012</td><td>CLO3</td><td>LO1, LO2, LO3</td><td>4</td></tr>
</table>

3B	Find the enthalpy, specific volume of steam at 12 bar and 350^0C.	**2**	CO3	PO1, PO2, PO3, P012	CLO3	LO1, LO2, LO3	**3**
3C	An air-conditioning system is to take in outdoor air at 10°C and 40 percent relative humidity at a steady rate of 45 m^3/min and to condition it to 25°C and 70 percent relative humidity. The outdoor air is first heated to 20°C in the heating section and then humidified by the injection of hot steam in the humidifying section. Assuming the entire process takes place at a pressure of 101.3 kPa, determine (a) the rate of heat supply in the heating section and (b) Change in specific humidity of the air. Show the process in a psychrometric chart	**4**	CO4	PO1, PO2, PO3, PO6, PO7, P012	CLO4	LO1, LO2, LO3, LO5, LO7	**3**
4A	Air enters a window air conditioner at 1 atm, 30°C, and 80 percent relative humidity at a rate of 10 m^3/min, and it leaves as saturated air at 14°C. Show the process in a psychrometric chart. Find out the heat removal rate in watts and amount of water contents removed in grams.	**5**	CO4	PO1, PO2, PO3, PO6, PO7, P012	CLO4	LO1, LO2, LO3, LO5, LO7	**3**
4B	Draw a plot relating compressibility factor, reduced pressure and reduced temperature. What are the key understandings from the chart?	**3**	CO4	PO1, PO2, PO3, PO6, PO7, P012	CLO4	LO1, LO2, LO3, LO5, LO7	**2**
4C	What is compressibility factor? What is the value of compressibility factor for an ideal gas?	**2**	CO4	PO1, PO2, PO3, PO6, PO7, P012	CLO4	LO1, LO2, LO3, LO5, LO7	**2**
5A	A gas power cycle working on Diesel cycle has compression ratio of 16. The isentropic compression	**5**	CO5	PO1, PO2,	CLO5	LO1, LO2, LO3,	**4**

	starts at 15^0C and 100kPa pressure. Heat addition takes place until the temperature raises to 1480^0C. Draw a neat P-v and T-s diagram and calculate the following. a) The cut off ratio b) The heat added in kJ/kg c) Efficiency of the cycle d) Mean effective pressure Given values: Gas constant, R= 0.287 kJ/kg-K, C_p=1.005 kJ/kg-K C_v=0.718 kJ/kg-K			PO3, PO6, PO7, P012		LO5, LO7	
5B	P-v diagram of a gas power cycle is shown in the figure 2 below. The specific volume at the beginning of compression is 0.9 m^3/kg and that at the end of compression is 0.06 m^3/kg . Cut off ratio is 2.1. Maximum temperature and maximum pressure of the cycle is 2560^0C and 6MPa. If total heat added in cycle is 1675 kJ/kg, find out how much heat is added in two separate heat addition process. Given values: Gas constant, R= 0.287 kJ/kg-K, C_p=1.005 kJ/kg-K C_v=0.718 kJ/kg-K . Compression starts at 0.1MPa and 40^0C.  \ **FIGURE 2**	**3**	CO5	PO1, PO2, PO3, PO6, PO7, P012	CLO5	LO1, LO2, LO3, LO5, LO7	**4**
5C	P-v diagram of a gas power cycle is shown in the figure 3 below. Calculate the efficiency of the Cycle.	**2**	CO5	PO1, PO2, PO3,	CLO5	LO1, LO2, LO3,	**3**

	Given: Specific heat of working fluid at constant pressure and constant volume are 1 kJ/kg-K and 0.66 kJ/kg-K respectively.  **FIGURE 3**			PO6, PO7, P012		LO5, LO7	

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***10***
2	***9***
3	***6***
4	***15***
5	***10***

BL	Total Marks
2	09
3	22
4	19
5	00